



AGUSAN DEL SUR STATE COLLEGE OF AGRICULTURE AND TECHNOLOGY

**OPTIMIZING PRINTING AND SCANNING COST
USING PIXEL-BASED ALGORITHMS**

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BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

FEBRUARY 2025



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FEBRUARY 2025

TRANSMITTAL SHEET

This undergraduate capstone project thesis outline entitled, **“OPTIMIZING PRINTING AND SCANNING COST USING PIXEL-BASED ALGORITHMS”**, prepared and submitted by **ARMANDO V. LORO JR.** and **FERNANDO O. GERALDEZ**, in partial fulfillment of the requirements for the degree of **BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY** is hereby accepted and endorsed.

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Chapter I

INTRODUCTION

Background of the Study

Printing and copying of papers are routine functions in the majority of industries, example, education, business, and government. Their cost is usually over or under estimated and thus wastage of funds and unnecessary spending occur. Traditional cost estimating is typically fixed price contract that ignores the actual composition of the document, example, text-to-image ratio. By applying a Pixel-Based Algorithm, organizations can obtain an improved and more realistic cost estimation system to optimize their utilization of resources.

Some research has shown technological advancements concentrating on cost estimation and print efficiency. Use the "Insert Citation" button to add citations to this document.

Use the "Insert Citation" button to add citations to this document.

talk about the advantages of Aerosol Jet Printing (AJP) in achieving high-resolution print on personal electronic devices, but there are disadvantages with controlled electrical performance only. (Montgomery et al., 2023) discuss the rapidly changing wave in additive manufacturing (AM), i.e., 3D printing, that integrates quality, material efficiency, and speed and hence becomes a stronger contender for commercialization. (James et al., 2022) present the growing need to verify image authenticity for forensic and legal applications, thus requiring advanced

image detection methods. (Lu et al., 2023) highlight the benefit of additive manufacturing (AM) over conventional manufacturing, where reduced cost is supported by facilitating resources and design freedom. (Krachmalnicoff & Tomasi, 2019) depict machine learning algorithms employed in handling big data and how they demonstrate that automated systems have been able to exhibit greater efficiency and decision-making. The study favors the use, cost reduction, and application of intelligent algorithms toward improving printing and scanning.

Even though printing and scanning technology has improved, most companies are still troubled with cost estimation. The cause is that a large number of users lack a good system of calculating the true cost of scanning and printing. High-density image documents, most of the time, use more ink, but pay the same as a text document. This results in overpayment and underpayment and hence cost inefficiency. Additionally, companies are left with unnecessary paper and ink usage due to poor cost control, resulting in unnecessary expense as well as environmental pollution. Without an intelligent automated system, end-users are unable to make intelligent decisions on document processing needs based on intelligent decisions.

Thus, in this research, it is suggested that a Printing and Scanning Cost Estimation System be developed using a Pixel-Based Algorithm. The system will scan documents by verifying text and image coverage to determine the actual cost of scanning and printing. With automatic estimation, the decision-makers will be able to make decisions based on facts, thereby eliminating unnecessary print costs and utilizing ink and paper to the fullest. The system will also give real-time cost analysis, thereby allowing organizations to implement cost-saving scan and print procedures.

Lastly, the use of a Pixel-Based Algorithm for Estimation will greatly improve the productivity and accuracy of scanning and print procedures. By eliminating inefficiencies of losing cost estimation and of wastage of resources today, the system will ensure a reliable and effective solution for users. The objective of this research is to bridge the gap between current cost estimation technology and technologies-based cost estimations such that an effective cost-saving document process system is guaranteed.

Statement of the Problem

While demand for effective document processing is on the rise, all are unable to estimate the cost of printing and scanning. Traditional price policies include pre-determined prices irrespective of the contents of any document, and therefore, there occurs underpayment and overpayment. All resources get wasted wastefully, and unjustified money wastage is incurred without any system of cost estimation.

The research seeks to identify and address the following problems:

- **Mis-estimation of Costs with Fixed Pricing Models** - Existing pricing models for printing and scanning do not consider actual content in a document, e.g., text-to-image ratio or color intensity. This leads to discriminatory pricing, plain text documents being overcharged, or image documents being undercharged, resulting in cost inefficiencies.
- **Automated Failure to Distinguish Black & White and Color Documents**
 - Most printing systems never distinguish between black & white and color

documents. This leads to mispricing, where color documents are priced as black & white and black & white documents are charged as colored documents. Organizations lack cost fairness and transparency when there is no classification system that automatically classifies documents.

- **Inefficient and Time-Consuming Cost Estimation Process** - The cost estimation process is cumbersome and error-prone when executed manually. The users cannot estimate the cost of scanning and printing and thus pay unwanted fees. A Pixel-Based Algorithm-integrated Optical Character Recognition technology must be applied to make the cost estimation process efficient and effective.

Objectives of the Study

The overall goal of this study is to design and develop a Printing and Scanning Cost Estimation Application. More specifically, the goal of this study is:

- **Integrate a Pixel-Based Algorithm for Cost Estimation** - Develop a sophisticated algorithm that accounts for the text and image coverage area of a document in order to estimate true printing and scanning costs.
- **Auto Document Identification and Black & White and Color Classifications** - to be priced correspondingly because color printing would be pricier in most cases than black & white.
- **Automatic Print and Scan Cost Calculation** - This technology calculates print and scan costs precisely using pixel-based formulas and optical

character recognition algorithms. It transforms the image-based and text-covered document, ensuring an equitable and fair price display.

Significance of the Study

The research discussed above is essential to transforming cost management in organizations that deal with gigantic volumes of documents. Compared to the limitations of classical flat-rate cost estimating, pixel-based algorithms, which constitute the crux of the research, are more precise and faster methods of isolating color from non-color content. This leads to efficient cost allocation, maximum resource utilization, and overall cost reduction.

Businesses and organizations. can gain an advantage from the research findings by reducing the cost incurred for scanning and printing, increasing overall resource management, and being cost-effective with pixel-based cost estimation.

IT and Print Management Professionals. The research offers evidence-based suggestions that can be used to improve existing document processing systems such that professionals can implement more accurate resource management and billing procedures.

Researchers and Scholars. Research for this paper provides a template for the design of pixel-based cost estimation algorithms, and it is calling for additional research in cost estimation in document processing and new areas of research.

Environmental Sustainability Benefactors. By minimizing unwanted printing and streamlining efficiency, the paper indirectly promotes sustainability by reducing the

wastage of paper and ink. The research discussed above is essential to transform cost management in organizations that deal with gigantic volumes of documents. Compared to the limitations of classical flat-rate cost estimating, pixel-based algorithms constituting the crux of the study are more precise and faster methods of isolating color from non-color content. This leads to efficient cost allocation, maximum utilization of resources, and overall cost reduction.

Scope and Delimitation of the Study

Scope

The proposed work is to maximize printing and scanning expenses based on pixel algorithms to improve the estimation of document content and estimate real-world samples at a reduced cost. The system uses significant functionalities like scanning, pixel-by-pixel analysis, real-time cost estimation, and reporting. The system is suggested to help small and medium businesses (SMEs) minimize print and scan expenses by classifying color and non-color content, monitoring resource use, and estimating the exact cost per process operation of each document process.

The system can:

Document Processing Capabilities

- It can upload and scan documents for processing.
- It can be processed by pixel for computing color use.

- It can estimate real-time print costs based on perceived color intensity.
- It can produce cost reports to track expenses.
- It can process print jobs with the cost-saving feature.

Admin

- It can add, change, and remove accounts (system users).
- Cost configurations can be configured to print and scan.
- It can print and email cost reports.
- It can display system activity logs.

Users

- It can scan and upload files.
- It can display the approximate cost of printing.
- It can choose cost-saving print choices.
- It can print your cost reports locally.

Location

Agusan del Sur State College of Agriculture and Technology (ASSCAT), San Teodoro, Bunawan, Agusan del Sur.

Delimitation

The study limitations are as follows:

- There is no interconnectivity with third-party accounting applications, therefore not for complete financial management purposes.
- It accepts standard document input only, such as PDF, PNG, and JPEG, but not less commonly used file extensions.
- It is limited to pixel-based color detection and lacks high-end image processing algorithms like optical character recognition (OCR) or pattern recognition.
- It is designed for small and medium-sized businesses, not large commercial print companies needing high-end industrial applications.

Time and Place of the Study

The research will be conducted in San Teodoro, Bunawan Agusan del Sur (ASSCAT). The machine will be fabricated at Agusan del Sur State College of Agriculture and Technology (ASSCAT). Machine testing will be commenced in March 2025 and end in May 2025.

Conceptual Framework

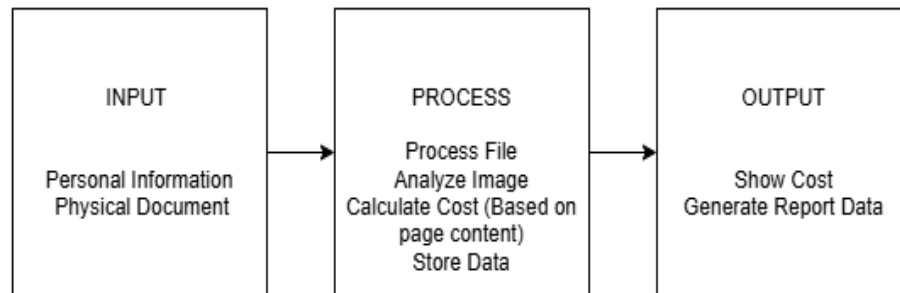


Figure 1 Conceptual Framework of the Study

Conceptual framework describes the application of pixel-based algorithms to be capable of optimizing scanning and printing costs. Its three components are Input, Process, and Output.

Operational Definition of Terms

Pixel-Based Algorithm - A software program that searches for individual pixels in a document to determine the ratio of color to black and white data to calculate the exact cost of printing and scanning.

Cost Optimization Software - Software that calculates the cost of printing and scanning by costing based on resource usage, i.e., color saturation and pages.

Document Cost Estimation - is the technique for determining the actual cost of printing or scanning a document based on its color distribution, page count, and usage of printer resources.

Color Ratio Analysis - A technique applied to identify the ratio of the number of colored pixels and black-and-white pixels, defining the overall cost of printing.

Print Job Management - Scheduling and managing print jobs on documents to reduce expenditure and enhance efficiency, e.g., priority and cost-saving printability.

Real-Time Cost Calculation - Real-time automatic calculation when scanning or uploading a document with the facility to provide instant feedback of estimated cost before printing.

Scanning Effectiveness - The ability of the system to scan and read documents in document analysis speed and pixel-based detection accuracy.

Resource Use Data - Utilize paper, toner, and ink use amounts utilized on paper in measuring papers for optimum cost utilization and wastage minimization.

Cost Report Generation - The ability to automatically create comprehensive printing and scanning cost reports and supply usage trend and cost-saving data.

Print Process Optimization - Printing the ideal process strategically without wasting spend and enhancing productivity with data-driven and real-time decisions.

Chapter II

REVIEW OF RELATED LITERATURE

Pillow Library

Pillow (PIL) is a Python image library. It is used in the script to impose logos on generated QR codes and merge 59 images. The color layers for single colors (red, green, and blue) are addressable through image processing when decoding. The library `qrcode[pil]` is required to build QR codes (number 61). It provides the potential to build QR codes to be used. The QRGB code is obtained by mixing 62 colors (red, green, and blue). It also provides 63 options to change other QR code parameters and error correction levels (Anderson, 2024).

Python package named Pillow which is capable of handling images. PIL was continued by Pillow. PIL existed from 1995 to 2009 and was originally developed by Fredrik Lundh. When Lundh stopped working on PIL, the project was inherited by Alex Clark. Clark started working on Pillow using the remaining work on PIL and forked it into its own project. Pillow supposedly focuses on marketing and helping PIL's ongoing evolution through ongoing releases, public evolution, and constant integration testing, Clark says (Harmaakivi,2023).

OpenCV Library

"Face Detection and Recognition Using OpenCV" examines 20 articles from 2018 to 2020 on the use of OpenCV in visual computing, security, public attention

automation, and traffic control. It sets its cross-cutting significance in delinquency detection, security, and traffic control. The article quotes gaps in current literature in that few applications are carried out in unregulated settings, no comparative study, real-time processing, and ethical and privacy concerns. It has as its mission enhancing awareness and use of OpenCV in the industrial society (Hasan & Sallow, 2021).

The fishing industry utilizes mobile technology to enhance quality and price of fish. The process is simplified, and efficiency of work improved with the use of an Android program, which recognizes and scans fish features through image processing. Despite this, there are performance focus problems and power consumption issues that exist in applications. To remove them from such issues and push the industry forward, the project seeks to create an efficient and user-friendly application using the OpenCV library (Qashlim et al., 2020).

Facial recognition methods such as Eigenface and Fisherface are implemented in most sectors to a large degree. However, there is an issue of how effective they would be when implemented using different libraries. With the empirical study on the effectiveness of OpenCV and Scikit-learn libraries in implementing these methods, this study seeks to bridge this gap and present informative findings to scholars and practitioners (Aliyu et al., 2022).

Optical Character Recognition (OCR)

"Survey of Post-OCR Processing Strategies" is concerned with improving OCR-acquired text quality for ancient texts. Noisiness in machine text slows down natural language processing and information retrieval. Similar post-OCR processing

methodology analogies by heterogeneous datasets and measures create research lacunas. The paper attempts to fill lacunas through existing methodologies integration, explaining trends, and giving directions to future work (Nguyen et al., 2021).

Though discovered to be of maximum use in document processing and data extraction, optical character recognition or OCR is not that effective in special character recognition. For general characters, the aforementioned 95% level has been discovered to be true as per studies, but for special characters, there is no fluctuation in the error levels. Context-sensitive approach is becoming highly popular for improving accuracy in the case of general characters. There is a gap here, especially considering the greater EasyOCR bias. The objective of this study is to close such gaps and position OCR technology as a legitimate field of research (Salehudin et al., 2023).

Scanning Cost Efficiency Techniques

Line scanning technology and collaborative robots such as the UR5 can be utilized to upgrade the manufacturing sector. This is due to the fact that human operators are usually interacting with the conventional inspection systems and hence leading to inefficiency and increased cost. The automation is likely to improve inspection attributes and decrease the cost. The challenges such as precise scans on reflective surfaces and communication effectiveness between the robot and the scanning system must be studied in order to develop solutions. Further research must be undertaken on wireless communication technology, scalability, and inter-

transferability of automated inspection technology across various manufacturing environments (Lopez-Hawa et al., 2019).

3D scanning technology, its increasing use in diagnostics for precision and custom-made medical implants. These aside, some caution in the literature is raised with respect to most particularly the process of integrating 3D scanning into the practice's activities and cross-validation across specialty domains. It is believed that 6% of the total studies of 3D scanning are in medicine and thus a need is conjectured to extend its use to cross-fascial application (Haleem & Javaid, 2019).

a cost-effective technique for scanning electrochemical microscopy and scanning ion conductance microscopy simultaneously, to overcome surface characterization problems in topography and electrochemistry. Economical production is proposed by the authors through the use of easily accessible materials, hence the processes become feasible. Issues of long-term stability, reproducibility, and performance of the probes with different types of samples still persist. The research centers on the establishment of more uniform and universal non-contact surface test equipment to promote basic studies and practice applications (Wert et al., 2020).

Pixel-Based Algorithms Cost Optimization

Edge detection with pixel-based matching and contour mapping method integration for the aim of enhancing performance. The research utilizes mask-propagation techniques as well as a new module of pixel-matching in order to enhance the matching between the target and reference pixels. The system is processed faster compared to normal processing and is more accurate, thus

enhancing the likelihood of detection and the reduction of the identification time. The paper utilizes sophisticated methods such as channel attention and non-local structures to enhance the efficiency of feature extraction and processing (Arulananth et al., 2023).

The literature of building footprint extraction has been discovered to have various methods and results. Existing studies have improved precision using image processing, such as edge detection and pixel value analysis. However, these studies are inclined to use idealized urban landscapes, thus opening the door to inapplicability. This research, with the aid of advanced machine learning techniques such as LightGBM, XGBoost, and Neural Networks, integrates the pixel-level as well as object-level segmenting procedures for attaining a total accuracy rate of 99.39% and an F1 score of 0.9935 for providing a more improved solution towards urban footprint mapping (Abujayyab et al., 2023).

"Comparing Object-Based and Pixel-Based Methods for Local Climate Zones Mapping with Multi-Source Data at 50 m and 100 m Resolution" compares object-based and pixel-based methods in Local Climate Zones (LCZ) mapping using multi-source data at 50 m and 100 m resolution. Object-based methods perform better than pixel-based methods in per-class accuracy and OA, with 2% higher OA at 50 m and 5% higher OA at 100 m. The research stresses the importance of the improved accuracy and efficiency of LCZ mapping methods (Yan et al., 2022).

Image Processing for Cost Reduction

"A Review of Deep Transfer Learning and Recent Advances" points to the increasing applicability of DTL in solving problems in healthcare and manufacturing industries, e.g., vast quantities of labeled data and costly training. Pre-trained models can decrease training time and resources, enabling AI-based systems on edge devices, it proposes. Nevertheless, the paper also identifies primary gaps like catastrophic forgetting and biased pre-trained models, and invokes tailored DTL solutions for universal use (Iman et al., 2023).

Efficient Super-Resolution Transformer (ESRT) is proposed as a method to improve deep learning feature extraction, particularly in fields like satellite imagery and medical imaging. It combines the Lightweight Transformer Backbone and the Lightweight CNN Backbone for learning long-term dependencies in an image. Nevertheless, the paper underscores the research requirement for optimizing the Transformer architecture in SISR under the tradeoff between performance and computational efficiency. This review is aimed at promoting the application of super-resolution techniques in all fields (Lu et al., 2023).

Grayscale Conversion for Ink Reduction

The Invention of digital light processing (DLP) for enhancing print resolution by pixel-level grayscale control. DLP is precise and efficient, but constraints such as overcuring, pixel shape constraint, and stair-step effect make it impossible to build fine details and smooth surfaces. Researchers introduce a reaction-diffusion model to get rid of the aforementioned constraints, providing a

systematic plan for DLP printing. The paper requires additional research to overcome existing constraints in DLP printing (Montgomery et al., 2023).

The study "Full Color and Grayscale Painting with 3D Printed Low-Index Nanopillars" examines the mechanism of how two-photon polymerization (TPL) is utilized to create nanoscale features in an effort to provide accurate color and grayscale painting. The authors believe that the mechanism can render the products sustainable and durable but also mention the limitations such as color accuracy and inability to offer a broad grayscale range. The research also proposes further study to develop color generation techniques in other areas such as electronics, design, and art (Wang et al., 2021).

Research "Grayification a useful grayscale conversion to facilitate handwritten historical document analysis" addresses image problems of documents, particularly in document manuscript processing. It argues that grayification improves preprocessing techniques and normalizes contrast for foreground and background objects of documents. However, there are some arguments in the literature that are not confirmed about whether grayification with different types of documents and conditions is useful or not. It also outlines the needs for additional investigation of how grayification incorporation can be integrated into current procedures for layout analysis and OCR (Bouillon et al., 2019).

Adaptive Thresholding in Scanning

adaptive thresholding method for robot and computer vision point cloud registration. The method complements the process by considering system kinematics and data type, further improving odometry performance in all fields of use. Questions do, however, require answers in terms of efficiency of these methods under various types of sensors and operating conditions. The method puts forward an attempt to bridge this gap and pave the way for more research in adaptive algorithms (Vizzo et al., 2022).

This is a new medical imaging method with adaptive thresholding varied based on local image properties, especially in tumors for PET/CT scans, which has proven to be very useful in segmenting tumors. It is highly appropriate in DLBCL where measurements using MTV would be nearly impossible due to the high heterogeneity of the tumors. Adaptive thresholding has been applied on rare occasions only to lymphoma imaging, and no established evidence of utility exists. Closing these knowledge gaps would improve tumor detection, treatment planning, and prognosis classification (Barrington et al., 2021).

Segmentation is an important operation in computer image processing, which divides images into homogenous sub-regions according to features such as color or texture. Techniques like multilayer thresholding, swarm optimization, differential evolution, and genetic algorithms have been proposed to improve performance. Entropy-based techniques surpassed conventional techniques, and meta-heuristic algorithms like beta differential evolution (BDE) have been proposed to improve segmentation under adverse imaging conditions (Bhandari, 2020).

Edge Detection for Printing Cost Reduction

Computer vision edge detection has a bearing on applications including autonomous vehicles and medical imaging. Although machine learning is cheaper and faster, traditional methods are still dominant for computational efficiency vs. accuracy trade-offs. Hybrid algorithms based on machine learning and traditional ones can potentially be cheaper with improved performance. Yet, systematic investigation and application-oriented framework are not present. In cost-conscious industries, additional light-efficient neural network design and frameworks have to be developed (Jing et al., 2022).

JSENet is a novel semantic segmentation and semantic edge detection architecture for 3D point clouds that outperforms previous approaches. It overcomes challenges such as ground truth label noise and enhances segmentation accuracy and edge detection accuracy. The method is founded on joint learning method and dual semantic edge loss, resolving open problems in 3D scenes. This provides a foundation for extending research to tasks such as indoor and outdoor scene perception (Hu et al., 2020).

Pixel-Based Printing Optimization

Machine learning (ML) to be applied with 3D printing has mostly been print-process optimised but design-dependent and inefficient. R&D is increasingly being redirected to user-level solutions, e.g., GUIs, in an attempt to minimize trial-and-error and material loss. One possibility is to design generalisable ML models that

can be fine-tuned with varying CAD designs and print parameters through the introduction of a "complexity index (Rahmani Dabbagh et al., 2022).

Philippine studies acknowledge low cost, low volume manufacturing and highly skilled professionals' challenges which 3D print businesses such as 3D2Go and 3Dprint.ph encounter. But opportunities of mass customization and sustainability are enticing. 3D printing consumes 50-100 times more energy than traditional techniques, and hence power-saving techniques are imperative. Studies target filling skill gaps and reducing technology adoption barriers (*2019 IEEE 11th International Conference on Humanoid, Nanotechnology, Information Technology, Communication and Control, Environment, and Management (HNICEM)*, 2019).

Edge Detection for Print Optimization

Automatic Surface defect detection in concrete structures and 3-D printed goods. In addition to dimension and roughness, they are the key drivers of their quality. Conventional human inspection detection is far from ideal. Manual inspection is time-consuming, error-prone, and tends to result in loss of resources. Surface irregularities are identified via object detection based on deep learning and image processing applied by Google Cloud Machine Learning (ML) Engine. Two methods of image processing are presented in this study. Pixels are taken into consideration in both instances in order to distinguish between a smooth or rough surface and an image from a USB camera. The deep learning-based solution is referred to as MobileNet-SSD in which it has a base convolutional neural network

that can be employed as an image feature extractor, which is used together with Single Shot MultiBox Detector (SSD) as an object detector. The model was successfully trained with Google Cloud ML Engine on a dataset of more than 20,000 images (Garfo et al., 2020).

The analysis of an image typically entails the use of a segmentation method. An example segmentation technique is through finding the edges of an image. There are several edge detection algorithms that have been put forward by researchers. This systematic review of literature seeks, therefore, to discuss how researchers have compared the quality of methods in an effort to provide a gauge of how effective the edge detection algorithms are (Tariq et al., 2021).

Halftoning

"reversible halftoning," which employs a two-convolutional neural network structure to synthesize the original color version from halftones. It solves the problem of lost details and color when color images are stored as binary halftones. The technology brings new possibilities for digital printing and imaging technology, yet much work remains to be accomplished in order to strike a balance between restoration fidelity and visual quality for varying print conditions and incorporate these innovations harmoniously into processes in the printing industry (Xia et al., 2021).

The function of halftoning is to use pixels with restricted intensities to two significantly different levels in an effort to mimic a continuous-tone image. All

printing relies on the technique, and most use fast cures-ordered dithering and error diffusion-techniques that do not attempt to preserve subtle structure that degrades halftone images. Computational cost of other old techniques' visual enjoyment through the optimum halftone solution is a less desirable attribute (Jiang et al., 2023).

Since halftoning processes maintain image quality at a cheaper impression price, advertisement has extensive uses of halftoning. Halftoning of a digital image can generate a binary image formed by dots. Halftoning minimizes visual information in the output; it also conserves ink while printing graphics. The information not accessible to the human vision, but the printed image is of acceptable quality. Floyd-Steinberger Most common technique for halftoning uses a fixed matrix for the conversion of halftone. Most proposed techniques of halftoning use pre-set kernels that do not provide image adaptation (Ortega-Sánchez et al., 2020).

Image Segmentation for Printing Cost Savings

The research emphasizes the significance of surgical guides and anatomic models printed in 3D to enhance surgical efficiency, especially in orthopedic and maxillofacial surgeries. Time saved in the operating room ranges from 12 to 66 minutes, corresponding to an annual cost saving of between 19,384 and 518,358. However, the current research is mainly alluding to maxillofacial and orthopaedic specialties without any prospective data being present to validate findings and provide extensive usage in other medicine specialties for 3D printing (Ballard et al., 2020).

Several complex structures can be built with 3D concrete printing. It is impossible to construct overhanging buildings without the use of supports since the properties of the material when new make it difficult. A support framework is therefore needed wherein overhangs of new material can be suspended, which often gets removed in post-processing, to take full advantage of 3D concrete printing. This research indicates that a single type of building material can be used to print the main structure and support structure if printing conditions are adjusted. It was found that nozzle travel speed and material volume flow rate significantly influenced the solidity ratio, which is real to predicted filament diameters' ratio. A characterization gap distance test was also performed to identify the maximum distance the filament is capable of covering without a visible slump (Tay et al., 2019).

Acceleration Techniques and Evaluation on Multicore CPU, GPU, and FPGA for Image Processing and Super-resolution

To set the stage for scaling up performance levels for image processing, this research is tackling accelerating applications in several cores and FPGAs. It is countering efforts towards improving SR algorithms and proving how computationally demanding they are, mainly when working with images that have low resolutions. As a reference and benchmark to the most advanced SR algorithms, the Low-Complexity Statistical Edge-Adaptive Back-projected Interpolation (L-SEABI) algorithm is introduced and examined independently. Performance measures like absolute performance, input size, power usage, and cost of ownership

are quantified on various hardware platforms, such as multicore CPUs, GPUs, and FPGAs. Comparisons of power consumption and speed of the platforms are conducted, and suggestions are made for their practical application. The acceleration techniques of the research are also applied to the Disparity Stereo Correspondence algorithm and prove feasible. The literature review provides an entire picture of SR acceleration, focusing on cooperation between GPUs and FPGAs to accelerate SR with a focus on worthwhile prospects and challenges (Georgis et al., 2019).

Energy Efficient Tag Identification Algorithms for RFID: Survey, Motivation and New Design

Two classes of anti-collision algorithms are used with RFID, namely Aloha-based and tree-based. The former utilizes random time slot selection, while in the latter, tags acquire tags autonomously for determining the presence of a tag. Power consumption is critical if the scanners in an RFID system are powered by batteries, which has yet to be solved despite the focus being on the minimization of tag recognition time as well as throughput optimizations. For companies to implement sporadic recharge and periodical scanning, energy-saving technology needs to be employed. By minimizing tag collisions, certain studies—such as EAFAS—solve energy issues at the cost of computationally costly and non-functional algorithms with mobile readers. To allow the application of green RFID technology in future IoT, this study formulates a fair method to maximize system performance and energy efficiency (Su et al., 2019).

An Algorithm and a Program for Configuring Risograph Printing, Taking into Account the Printing Properties

Risograph printing ensures the printing quality, affecting sharpness in an image and minimizing mechanical wear and tear on equipment. Black-and-white publication and book printing utilize offset paper with a 60–80 g/m² weight; the level of fibers determines stability and density in size. Risograph printing cannot deliver good prints, particularly with generic drivers, and thus leads to problems such as uneven reproduction and disappointing sharpness. Print quality is enhanced through the use of parameter tuning and high-quality rasterization. Cost is also a consideration; recycled paper is less expensive but less durable. Electronic filtering and risograph condition setting are some methods of enhancing print quality. Even after research, adjusting optimal paper quality and the influence of different ink content remains problematic. In addition to this, mechanical restrictions brought about by rigid paper still need to be eliminated. More research will be required to establish risographic printing and realize the cost-quality balance (Yudenkov & Sulim, 2024).

Cumulative Inaccuracies in Implementation of Additive Manufacturing Through Medical Imaging, 3D Thresholding, and 3D Modeling: A Case Study for an End-Use Implant

Individual additive manufacturing (AM) design to produce craniomaxillofacial implants individually for each patient is becoming more

prevalent. Individual process stages, such as 3D modeling or medical imaging, are less defined within the literature than end-to-end processes. Each step in implant manufacture, which could cause defects potentially influencing surgical planning, fit of the implant, and patient healing, is circumvented with this research limitation. Quality assurance is critical in medical imaging and AM manufacturing. A new QA Phantom is developed to check imaging test systems by applying metrology standards to check the precision of implants. However, error compounding within the process stream is not a significant issue of existing fabrication methods, resulting in improper implant installation. Enhanced procedures considering error calculation can provide maximum accuracy and enhanced surgical results. Although there has been research on individual aspects of AM, studies on cumulative errors in total, especially in CT imaging procedures and 3D thresholding, are very few. By investigating the impact of these variations on end implant accuracy and improving quality control methods for AM in medicine, our study will bridge this gap (Akmat et al., 2020).

Numpy Library

The Central Python library for array computation is referred to as NumPy. It is important to learn analysis pipelines in an assortment of areas such as materials science, physics, chemistry, astronomy, geoscience, biology, and psychology. economics, science, engineering, and finance. For example, NumPy was an indispensable component of the software stack implemented in astrophysics for the

very first imaging of a black hole² and the observation of gravitational waves¹ (Harris et al., 2020).

High Performance Computing (HPC) has been very much boosted by Python, already a dominant scientific computing language. NumPy, an effective implementation of the multi-dimensional array (tensor) data structure, and some fundamental arithmetic and linear algebra, is the basis for Python's platform. Python and NumPy's relatively inferior performance compared to the rest of HPC languages has created an enormous interest in separating the Python's expressive syntax from the implementation itself through compilers and frameworks (Ziogas et al., 2021).

A Proposal of Printed Table Digitization Algorithm with Image Processing

With an aim to enhance heart health recording and observation, especially in the case of past records in historical documents, the research attempts to set up an autonomous method of digitization that would convert paper scan ECG records into digital form. The method employed was advanced image processing techniques such as bit plane slicing and denoising, which were confirmed using Pearson correlation coefficients, repeatability, and measurement reproducibility. The mean of correlation was 0.94, and measurement error was below 1 mm, that is, there was high precision. Based on the outcomes, the algorithm provides a reliable means of keeping ECG data and facilitating timely diagnosis in clinical practice. Whereas this one and your research are both intended to increase access to ECG data and accuracy with newer algorithms, your research differs in that it's focused on signal quality and

print and scan cost savings, and both meet the clinical and economic objectives, and presents a different twist in healthcare solutions (Randazzo et al., 2022).

Advancements in 3D Printing: Directed Energy Deposition Technique, Defects Analysis, and Quality Monitoring

Quantitative evidence of how scan speed and energy density affect print quality has been confirmed by quantitative research on 3D printing, which also demonstrates that suitable scan speeds can optimize component integrity and melt pool behavior. Also, organized data approaches to deposition path planning have been considered within the spatiotemporal processing of Directed Energy Deposition (DED) to improve thermal distribution prediction and minimize computation requirements. Besides, machine vision technologies and real-time inspection are highlighted in the literature on flaw detection and quality inspection as central steps in material loss minimization and flaw detection optimization. Although the above study is on individual print processes, this project aims to utilize pixel-level algorithms in an end-to-end scanning and printing approach, minimizing expenses. This work universalizes these methods to make imaging and additive printing more cost-saving and efficient (Imran et al., 2024).

Document Image Quality Assessment: A Survey

In addition, programmed data solutions for deposition path planning have been considered in spatiotemporal processing of Directed Energy Deposition (DED) for better prediction of thermal distribution and fewer computation demands. In addition, machine vision technologies and real-time inspection are emphasized in flaw detection and quality inspection literature as key actions toward reducing material loss and increasing flaws. Single Document Image Quality Assessment (DIQA) tests, which use pixel-based algorithms to scan and print at low rates, are highly emphasized. Most methods employ optical character recognition (OCR) as a measure of quality quantification, whereas DIQA tests the readability and distortion effect of the document. Visible quality is not always paired with OCR-based scores. Two of the shortcomings of DIQA are human opinion rating for small datasets and the unfeasibility of real-time subjective scoring. Besides, document digitization processes have considered the effect of expensive cameras and camera phones on image quality regarding transmission distortion and compression artifacts. To make it more efficient and cut down on scanning and printing costs, efforts have been made to employ pixel-level techniques, but efforts have mainly focused on document quality estimation. The image quality of the document will be enhanced in this work with minimal resource consumption by embracing future technologies and improving current research methods Detection (Alaei et al., 2023).

Compression Techniques for Printing Efficiency

ANN research to minimize the lattice infill patterns for optimizing used artificial neural networks to target material efficiency in additive manufacturing.

This action saved output print 2.5 times per spool and cut material waste by 61.3%. Like grid patterns, variable infill line spacing can provide low-cost printing without compromising structural integrity, according to a study of the efficiency of cubic infill patterns. A comparison of infill patterns also revealed a compromise between print time and the economy of material, where there is a need to balance performance and cost. Both projects share an interest in minimizing wastage and optimizing 3D printing efficiency, which is the same as scanning and printing-based pixel algorithm optimization, but whose individual specific interest is optimizing 3D printing infill patterns. It has been manufactured increasingly at the core of cost reduction, even though it has been relocated from the paradigm of analysis patterns to machine learning. The change provides solid ground for the construction of document processing technology in the coming years, with resource optimization and productivity remaining important factors (Alejandrino et al., 2020).

Chapter III

METHODOLOGY

Research Design

The Waterfall Model is the software development approach utilized in this research, based on a developmental research design. The project aims to design, develop, and test a cost-optimization software based on pixel-based algorithms to enhance cost estimation for printing and scanning. With its linear, step-by-step model, the waterfall model insists on a phase, requirement analysis, system design, implementation, testing, deployment, and maintenance completed before proceeding to the next one. An experiment and quantity-oriented approach in which system testing is conducted for precision determination, performance, pixel measurement analysis, print costs, and timeliness. Some data collection methods include accuracy validation system tests, pixel-based vs. classical cost estimation document checks for comparison, and surveys for user satisfaction to determine the usability and impact. It is followed by the rigorous approach through which a valid and efficient cost-optimization system is created that can be implemented in Small Businesses.

Software Process Model

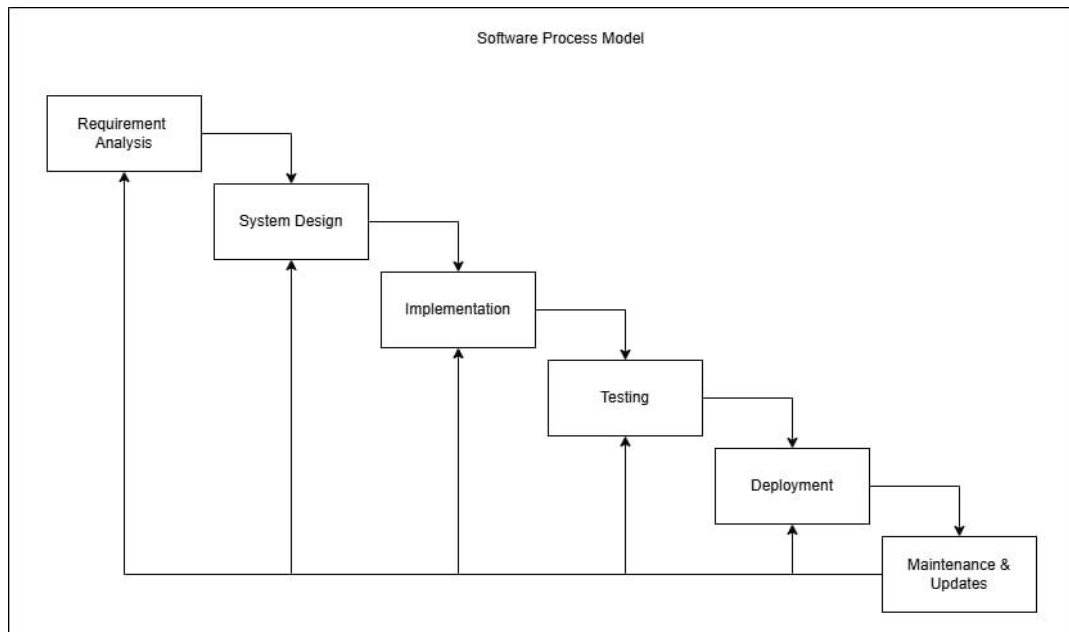


Figure 2. Phase System Development Life Cycle (Waterfall Model)

The Waterfall Model is depicted in the diagram as a sequential and linear, formal Software Development Life Cycle (SDLC) process. It consists of six phases: Requirement Analysis, in which system requirements are gathered; System Design, in which technical details like database schema and UI diagrams are designed; Implementation, in which development and coding are done; Testing, in which functionality and correctness are checked; Deployment, in which the system is released to end-users; and Maintenance & Updates, in which repairing and updates are done based on user grievances. This model will perform best on projects with known requirements, such as small business printing companies, which must follow a sequential development sequence.

Requirements Analysis

To reduce print costs by a pixel-based algorithm, the requirements of tiny printing houses need to be collected, analyzed and entered into requirements

specifications. Observation, interviews, and questionnaires are used to identify how print houses manage print costs and face challenges. The plan is to create a system that automatically computes costs from colors and shades in a document and proposes how costs can be reduced.

User Requirements

Administrative personnel, print business firms, and small organizations expect the system to:

- **Upload Documents:** Allowing the user to upload documents to validate should be possible.
- **Automated Cost Estimation:** The system should provide real-time printing cost estimations with the pixel-based algorithm.
- **Expense Reports:** Cost reports should be segregated on print charges and charges.
- **Cost-Saving Solutions:** The system should provide alternative printing means for cost-saving reasons.
- **Easy-to-use User Interface:** The system should be easy for a non-computer literate person.
- **Multiformat capability:** It should handle various document types, such as PNG, JPEG, and PDF.

Functional Requirements

Functional requirements specify the critical operations and characteristics to be executed by the system.

They are:

- **User Authentication:** Secure sign-up and login process.
- **Image Processing:** Reduce print and scan expenses by pixel-based algorithms.
- **Data Storage & Retrieval:** Upload and save user documents into MySQL.
- **Report Generation:** Generate printable reports out of processed information.
- **User Management:** Allow administrators to monitor user access and privilege.
- **API Integration:** Allow RESTful API endpoints for frontend-backend communication.

Operational Environment

Windows 11, Linux, or macOS offers the system web-based interface assistance through updated web browser assistance. The system is supported by a stable internet and an SQL database to save user data during cloud deployment. HTML and CSS offer the front end, and Django provides the back end.

System Interface

The system interface comprises:

- **Web-Based User Interface:** Supplied using a responsive design browser.
- **Admin Dashboard:** User management controls, setup files, and system settings.
- **Database Interface:** MySQL is used to store and retrieve structured information.
- **REST API Interface:** Frontend and backend data processing interaction.

Communication Interface

The system employs:

- **HTTP/HTTPS Protocols:** Secure client-server communication.
- **RESTful APIs:** Secure backend-to-frontend data transfer.
- **Email Notifications:** Integrate through SMTP for notification, confirmation, or system messages.

Software Interface

The system communicates with different software modules, such as:

- **Django Framework:** Handles backend operation and API services.
- **MySQL Database:** Reading/writing user data.
- **XAMPP:** To create a local development environment.
- **Pillow & OpenCV:** Image processing is done.
- **Django REST Framework:** Handling of the API endpoint.
- **Visual Studio Code:** Software development and debugging.

Hardware Interface

Below hardware is needed for the system:

- **Processor:** Intel Core i5 or above for smoother execution.
- **RAM:** 8GB at least (16GB for enhanced performance).
- **Storage:** Minimum 256GB SSD (512GB SSD for faster processing if needed).
- **Graphics:** Onboard graphics integrated into the motherboard or onboard in the event of more image processing.
- **Monitor:** Minimum 1080p resolution to provide comfort in UI.
- **Network:** Minimum 10 Mbps Internet for cloud-hosted access.

Non-functional specifications

- **Reliability:** The system must be 99% available with zero downtime during working hours.
- **Operability:** The system should be operatable and used by new users comfortably with a friendly interface.
- **Maintainability:** The system must be easy to change, debug, and maintain if future changes and patches are required.
- **Scalability:** It must support additional uploaded documents without being struck by compromised performance or efficiency.
- **Availability:** The machine must always be available during continuous operation under changing conditions.
- **Delivery:** Papers should be dealt with within five seconds to enable quick and efficient work.

System Requirements

Table 1. System Requirements

Component	Specification	Description
Operating System	Windows, Linux, and macOS.	Cross-platform deployment
Processor	AMD Ryzen 5 or Intel Core i5	supports effective data processing.
RAM	8GB RAM (16GB is preferred)	has enough memory to process multiple tasks
Storage	256GB SSD (preferably 512GB)	enhanced data access and read/write speeds.
Database	MySQL	holds all the system and user data.
Backend		Django's backend handles API calls and business logic.
Frontend	HTML and CSS	enable ease of use through an intuitive interface.
Libraries	Pillow, NumPy,	OpenCV, enhance computational speed and image processing.
Network	10 Mbps network	enables fast data exchange and cloud deployment.
Development Tool	Visual Studio Code	for coding and debugging.

Component	Specification	Description
Hosting		Cloud hosting for ease of accessibility and web deployment.

Software and Hardware Requirements

Software Requirements

Table 2. Software Requirements for Visual Studio

Software Tool	Description
Visual Studio	Visual Studio is a software development software developed by Microsoft. It is computer program development software with several programming languages like Python, JavaScript, C++, etc. Visual Studio has features like debugging, code editing, version control integration, and add-ins for enhancing productivity. Visual Studio is also supported for the front and back ends and can be termed an interactive environment to develop web, desktop, and mobile applications quickly.

Table 3. Python Software Requirements

Software Tool	Description
Python	Python is a low-level, high-level programming language that is easy to understand, readable, and flexible. It is also applied

extensively in web development, data science, automation, AI, etc.

Table 4. Django Software Requirements

Software Tool	Description
Django	Django is an open-source web framework written in Python that simplifies web application development by executing basic features such as authentication, database, and security. It employs the Model-View-Template (MVT) pattern and is appropriate for secure and scalable web application development.

Table 5. HTML, CSS Software Requirements

Software Tool	Description
HTML, CSS	HTML (Hypertext Markup Language): The default language marks web pages by assigning headings, paragraphs, images, and links. HTML offers the structure of a regular webpage. CSS (Cascading Style Sheets): A style language to specify the look and feel of a webpage, layout, color, font, or responsiveness. CSS is developed based on the style of HTML elements.

Table 6. XAMPP Software Requirements

Software Tool	Description
XAMPP	XAMPP is open-source, free software that installs the local server environment for web application development. It has a

	web server, Apache, a MySQL/MariaDB database, and server-side scripting languages, PHP and Perl. It is cross-platform and can be used on Windows, Linux, and macOS, making it a wonderful developer tool for developing, testing, and hosting websites locally before deployment.
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Table 7. Pillow, OpenCV, NumPy, Django REST Software Requirements

Software Tool	Description
Pillow, OpenCV, NumPy, Django REST	Pillow, OpenCV, NumPy, Django REST Pillow is a Python image-processing library that reads, modifies, and writes image files in multiple formats. OpenCV is an open-source library for computer vision that offers real-time image and video processing functionalities, which makes it the best option for image and video detection operations and facial recognition. NumPy is a library for numerical computing that supports large multi-dimensional arrays and mathematical functions required for data analysis. Django REST Framework (DRF) is a Django API building tool. It is meant to make developing sound, scalable, and secure API endpoints for web applications

easy.

Table 8. Windows 11 Software Requirements

Software Tool	Description
Windows 11	Windows 11 is Microsoft's new operating system with a fresh new, simplified, modern-looking interface, enhanced performance, and productivity functionality, a redesigned Start menu, richer multitasking functionality with

	Snap Layouts, better virtual desktops, and smoother integration with Microsoft Teams. Windows 11 has enhanced security, quick updates, and improved gaming performance using DirectX 12, Auto HDR, and Direct Storage. It is both business- and consumer-oriented and works with current hardware and software
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Table 9. Hosting Software Requirements

Software Tool	Description
Hosting	Hosting is the storage and maintenance of websites, applications, or databases on a server in a way that they are available via the Internet. It allows individuals and companies to put their content online. There exist various forms of hosting, such as shared hosting (multiple sites hosted on one server), VPS hosting (dedicated resources on a virtual private server), cloud hosting (scalable host with distributed resources), and dedicated hosting (single server for a single client). Hostinger, Bluehost, and AWS are the best-known hosting services that provide domain registration, security, backup, and performance optimization features to support smooth website performance.

Hardware Requirements

Table 10. Processor Hardware Requirements

Hardware Tool	Description
Processor	Intel Core i5 is an affordable processor that is efficient and has performance, thus making it perfect for tasks such as general computing. The processor has numerous cores and threads that enable multitasking without trembling, quick processing time, and increased power efficiency. i5 processors are typically employed in laptops and desktops to handle day-to-day applications such as internet surfing, office work, gaming, and even certain content creation activities. With onboard Intel graphics, it has entry-level graphics capabilities, with more expensive models having Turbo Boost technology for increased performance when necessary.

Table 11. RAM Hardware Requirements of Hardware Tool

Hardware Tool	Explanation
RAM	8GB RAM or more is essential for seamless system operation with the ability to open multiple programs simultaneously without delay. It improves multitasking, application launch of programs, and rapid execution of activities like browsing, document preparation, and simple programming. For applications like playing games, video editing, or advanced processing, the 16GB or higher upgrade is advisable.

Table 12. Hardware Requirements for Monitor

Hardware Tool	Description
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Monitor	The 1080p monitor with the minimum resolution provides clean and sharp graphics, which is ideal for coding, design, streaming video, and gameplay. The monitor ensures better readability, efficient workspace management, and a richer viewing experience. Even higher resolutions, such as 1440p or 4K, provide even more clarity and detail, primarily for professional graphic and editing purposes.
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Table 13. Hardware Requirements for Graphics

Hardware Tool	Description
Graphics	The onboard display or integrated graphics is the computer processor or motherboard's built-in graphics processing unit (GPU). Such graphics solutions are adequate for routine tasks like web surfing, video watching, programming, and light image editing. Though they are less capable than dedicated GPUs, they are power-friendly and ideal for systems that do not need heavy graphical processing, including business applications, programming, and database management.

Table 14. Network Hardware Requirements

Hardware Device	Description
Network	10 Mbps network connection provides a robust and solid internet speed for cloud-based services, online collaboration, and remote access. It facilitates smooth browsing, file uploads, and API

communications and can be ideal for web application hosting and medium data exchange. Nevertheless, where high-end processes such as file transfer or streaming video are required, an even faster connection might be needed.

Sampling Method

The study population will be small print businesses, including local photocopying companies, print shops, and computerized print businesses. It will include identified printing service clients, employees, and business owners.

The sampling method to be used will be stratified random sampling, which will enable a stratified sample of various types of printing businesses (i.e., offset, computerized printing, and photocopying companies).

The following will be taken into consideration while recruiting the participants:

- Should be working in small-sized printing businesses.
- Should know cost estimation and resource planning for printing.
- Fixed consumers of the printing service.

At least 30-50 respondents would be interviewed to have a representative sample.

Survey Instrument

A structured questionnaire will be utilized to gather more data regarding the problems, cost estimation procedures, and technology adoption in small print businesses. The survey will cover:

- Demographic Data (number of employees, location, size of the business)
- Printing Activities (Types of print services, prints per day, frequent problems)
- Cost Estimation Methods (Method used by companies to estimate the cost of printing, frequent errors)
- Use of Technology (awareness and use of cost estimation software)
- User Experience (the parts of a system most commonly used by users to determine the cost of printing).

Multiple-choice, Likert scale, and open-ended questions will be offered in the questionnaire for qualitative data.

Data Analysis

These will be used in data processing and data analysis of the data collected:

- Descriptive statistics for tabulation of the responses (percentages, means).
- Comparative analysis is the method of determining patterns among different business models.

Feasibility Issue

Economic Feasibility is comparing the estimated cost and benefit of the project to see if it is financially viable. The system uses pixel-based algorithms to lower ink and toner usage without affecting the document's readability and lower scanning and printing costs. Scan efficiency and print density management can save companies operational costs in the long run. Long-term savings are used in the development of software, hardware, and maintenance to make the project cost-effective for individuals and business corporations.

Technical Feasibility considers whether the current knowledge and technology are adequate to craft the system effectively. The project utilizes Python, Django, OpenCV, and Pillow for image processing, print density adjustment, and processing of scanned documents. A good and scalable database is required to hold its document content, have an effective hardware setup, and have adequate computation power. Image processing skills and algorithm design proficiency in the development team are also necessary to ensure smooth system operations based on the provided technology infrastructure.

Operational Feasibility ensures that the technology will run suitably when utilized and meet the user requirements. For an uncomplicated experience, pixel-based optimization should be implemented in installed scanners and printers without complications. Users should be able to manually optimize printed and scanned documents with minimal interference and with minimal clicks. It should allow integration with various devices and file types to make it easy for individuals and businesses to adopt without discontinuing their work.

Schedule Feasibility The Feasibility of the schedule determines whether or not the project can deliver performance and quality within a period that is not too long. Pixel optimization techniques must be created and enacted with much study, testing, and optimization for the best effect. Development must be timely enough to remain competitive and current in the market. Good project management must be applied so that deadlines will be met and a quality solution is obtained because delays in development will affect adoption levels or result in out-of-date technology.

Gantt Chart

Table 15. Gantt Chart

Task	Duration
Project Initiation	1 week
Planning & Research	2 weeks
Requirements Analysis	2 weeks
Algorithm Development	3 weeks
System Development	8 weeks
Testing & Optimization	2 weeks
Deployment & Maintenance	2 weeks

Work Breakdown Structure

Table 16. Work Breakdown

Level	Phase	Tasks
1	Project Initiation	Define project scope, objectives, and Feasibility
2	Planning & Research	Define objectives, gather requirements

3	Requirements Analysis	Determine system/hardware requirements and establish system requirements
4	Algorithm Development	Improve pixel-based cost algorithm
5	System Development	Develop backend (Django), frontend (HTML/CSS), and database (MySQL)
6	Testing & Optimization	Test the system, correct bugs, and enhance performance
7	Deployment & Maintenance	Deploy the system, document, and maintain.

Design Phase

The design phase focuses on organization, interaction, data flow, and process execution. To portray the system's running in a neat picture, a number of different kinds of graphics are used.

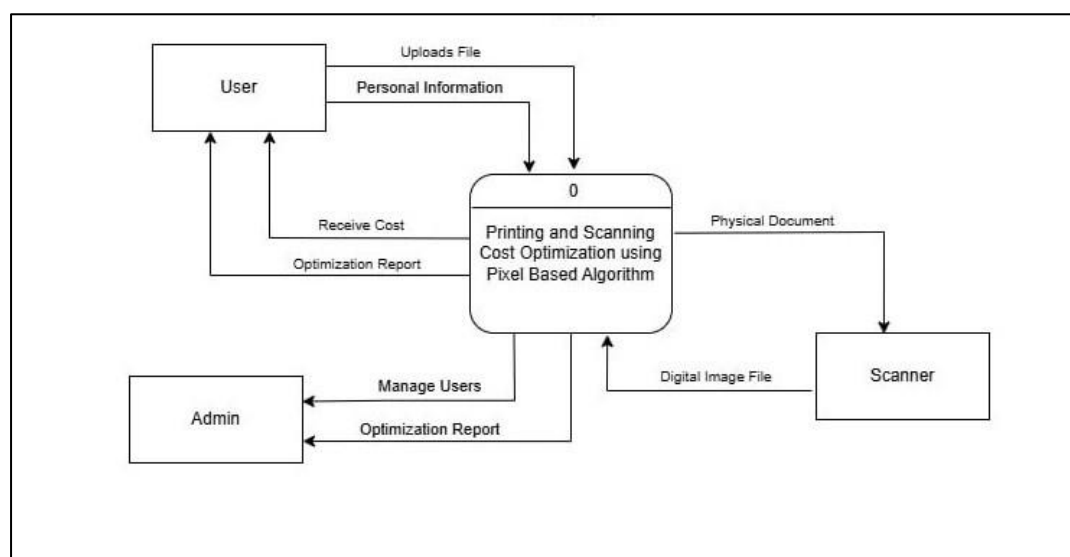


Figure 3. Context Diagram

The context diagram presents the print, scan cost optimization system and the outside world, including the user, scanner, admin, and high-level communication. When a file is loaded, and a user's information has been inputted,

the system optimizes the document, runs an optimization routine for costing, and displays the cost estimate and optimization report. The system also engages with a scanner by reading paper documents and accepting digital image files to process. The administrator can also read optimization reports and control users. The flow of data and significant interactions of the system are readily illustrated in this diagram.

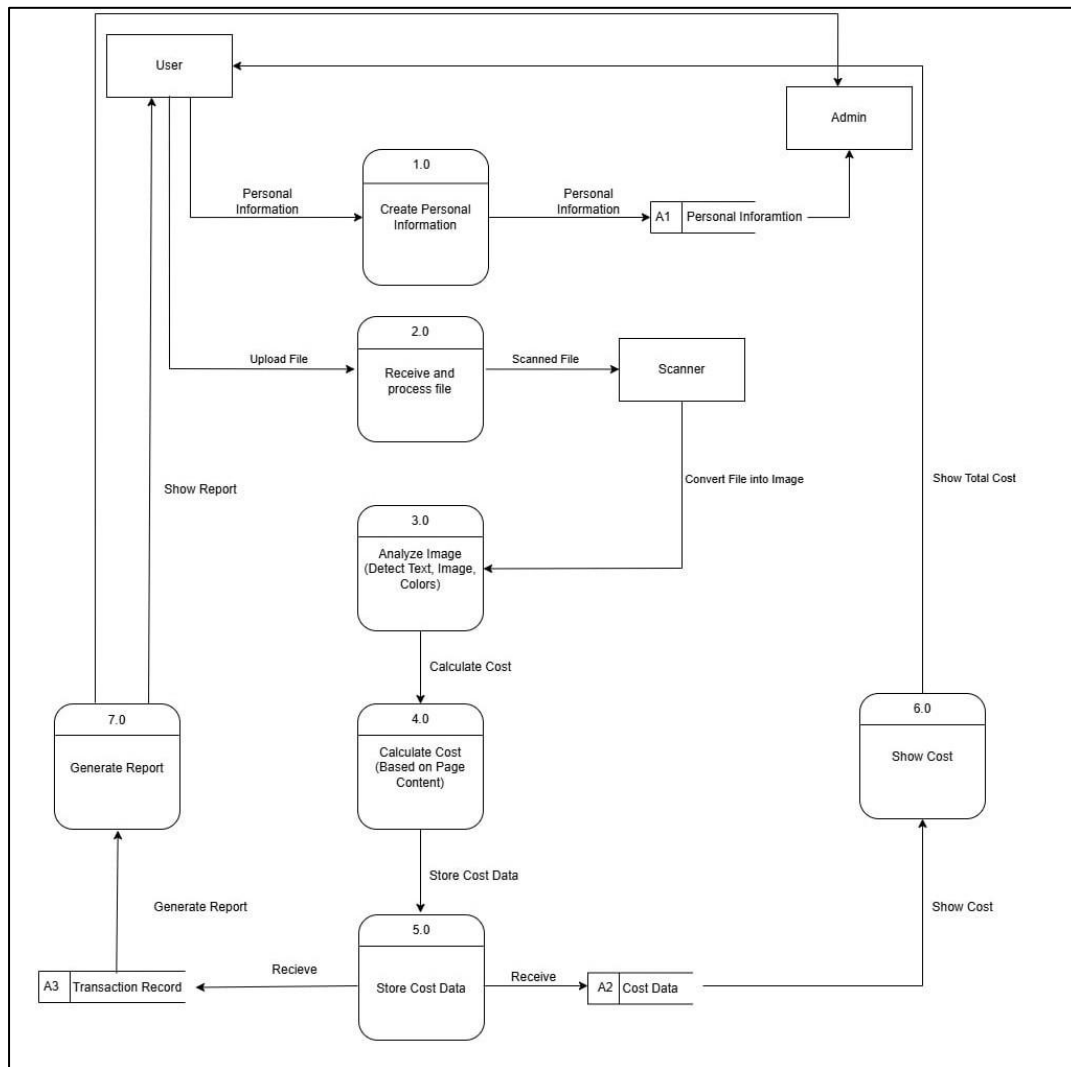


Figure 4. Data Flow Diagram

The figure above illustrates a file in this Data Flow Diagram (DFD). The file is processed, its content is checked, expenses are calculated, and reports are printed.

The administrator also utilizes the system to see expenses and personal information.

Significant elements of the DFD

1. External Parties:

- User: Uploads documents to be scanned and provides personal details.
- Admin: Views expense reports and user information.
- Scanner: Creates images to be processed from uploaded documents.

Data Flow:

- An administrator is given user-specific information.
- Once the user has uploaded the file, the file is scanned and converted into an image by a scanner.
- The cost is determined by the system reading the scanned image and identifying what it is.
- Upon saving, the cost data is retrieved so that it can be utilized for reporting.
- While the user is presented with a report generated, the administrator receives the complete expense.

Entity Relationship Diagram

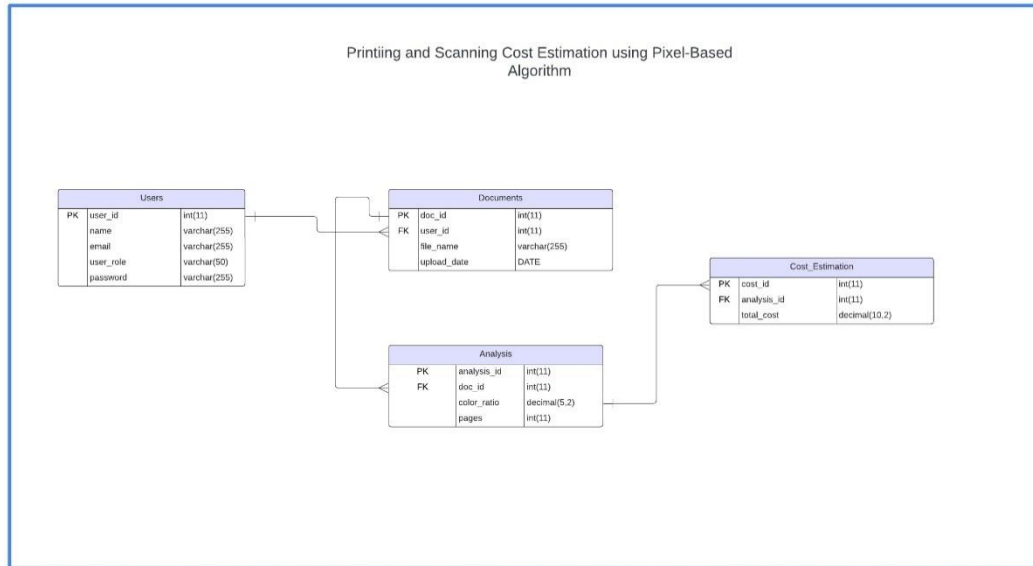


Figure 5. Entity Relationship Diagram

The diagram illustrates the Entity-Relationship Diagram (ERD) of an Optimizing Printing and Scanning Cost Estimation Using Pixel-Based Algorithms. There are four major tables: Users, to hold user information; Documents, to hold uploaded documents; Analysis, to analyze document properties such as color ratio and pages; and Cost_Estimation, to calculate the final cost after data analysis. All these entities are related to each other for efficient document processing, cost estimation, and user management.

Table 17. Entities & Relationships

Entity	Attributes
Users	user_id (PK), name, email, user_role, password
Documents	doc_id (PK), user_id (FK), file_name, upload_date
Analysis	analysis_id (PK), doc_id (FK), color_ratio, pages
Cost Estimation	cost_id (PK), analysis_id (FK), total_cost

Relationships: User Uploaded Documents → Documents are processed → Analysis matures Cost Estimation.

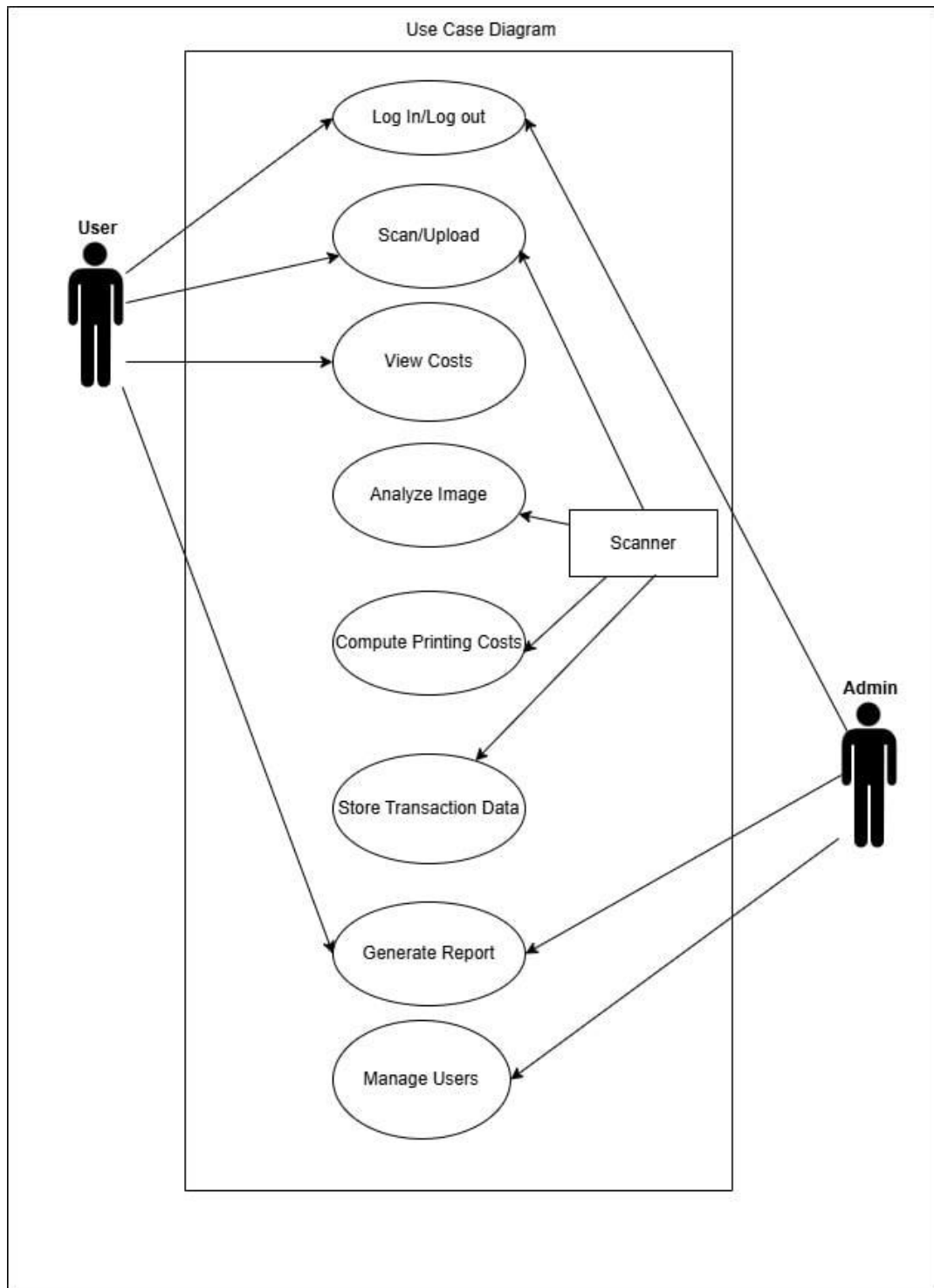


Figure 6. Use Case Diagram

The Use Case Diagram presents a graphical representation of how the users (Admin and User) use the Printing and Scanning Cost Optimization System.

The user can log in, upload or scan, view costs, utilize the scanner to scan images for examination, and calculate print cost estimates. The system stores computed

costs and transaction details. The administrator privileges provided are user administration and reporting. The images are processed in the cost aspect by integration with a scanner. The diagram helps describe the functionality of users and system functionality.

Table 18. Performers & Use Cases

Performers	Use Cases
Users	Upload Documents, View Cost Estimation, and Generate Reports.
Admin	Manage Users, Set Cost Calculation, and View System Logs.
System	Process Pixel-Based Analysis, Compute Costs, and Store Information.

Use Case Breakdown:

- Upload Document - The user uploads the document to the cost estimate.
- Perform Analysis - System processes document pixel by pixel.
- Generate Cost Estimation - The system calculates and shows cost data.
- Generate Report - The user can download the print cost report.

Project Implementation Plan

Project Cost

The approximate cost of the project Optimizing Printing and Scanning Cost Using Pixel-Based Algorithms is divided into development, hardware, software, and operating expense categories.

Table 19. Estimated Project Breakdown

Category	Description	Estimated Cost
Development Cost	Software Development and Coding	0
Software Tools	Python, Django, Pillow (Open Source)	0
Hosting & Domain	Domain	1500
Hardware	Computers or Laptops	15000
Testing & Debugging	User acceptance testing, bug fix	
Deployment	Server setup and initial launch	
Maintenance & Updates	Regular system updates, Bugs fixes, and support	
Training & Documentation	User manual, Training sessions	
Contingency Cost	Unexpected expenses	
Total Estimated Cost	One time	
System Maintenance		
Total Overall		

Risk Management

Risk management ensures the project stays on track by identifying potential risks and implementing mitigation strategies.

Table 20. Risk Management

Risk Factor	Potential Impact	Mitigation Strategy
Technical Issues	Bugs or software malfunctions	Regular testing and debugging
Hardware Failure	Printer/scanner breakdown	Use backup devices if available
User Adoption Resistance	Difficulty in system adaptation	Conduct training and provide user guides
Data Security Concerns	Unauthorized access to data	Implement role-based authentication
Budget Overruns	Increased development costs	Monitor expenses and adjust as needed
Project Delays	Late completion of milestones	Set clear deadlines & track progress

Code development, creation, and integration

There needs to be a development cycle to adhere to:

- System design by developing the database schema's entity relationship diagram (ERD).
- UI/UX design using wireframe tools like Figma.
- Programming: Python (Django) is employed to implement business logic for the backend.

- Frontend: User interactions are programmed with JavaScript, HTML, and CSS.
- Pillow (Python library) is used to estimate image processing costs.
- Compilation and Linking: Use pip to handle dependencies.
- Utilize Django ORM to communicate with the database.
- The front end and back end must be well integrated.

Organization and Quality Control

The following will be ensured to keep the quality of the code intact:

- Code Review: The team members of the coder must review the code before merging it into the re a branch.
- Unit testing: Get everything to function as it should function.
- Test the interaction between system components with each other by using integration testing.
- Code Documentation: Use well-documented code and code comments wherever needed.
- Version Control: Code changes will be pushed through GitHub/GitLab.

System Documentation

Detailed documentation of the following will be provided:

- The user guide provides step-by-step instructions on how to use the system.
- Technical documentation is system design, database schema, and API documentation.
- Installation Manual: Step-by-step installation guide for installing the system on different platforms.

System Installation and Configuration

Based on clients' requirements, the system would be installed locally or on cloud servers.

The installation steps include

- the installation of dependencies, i.e., Django and Python.
- Installation of application to integrate with databases (Xampp/MySQL).
- Installation and administration of static files are needed to maximize efficiency.
- Installation of access control and user authentication.

System Initialization and System Administration

At system installation, the following is done:

- First Testing: Testing if modules are working as desired.
- User Training: Documentation and training.
- Monitoring & Maintenance: Regular updates, performance tuning, security patching

.

Sequence Diagram

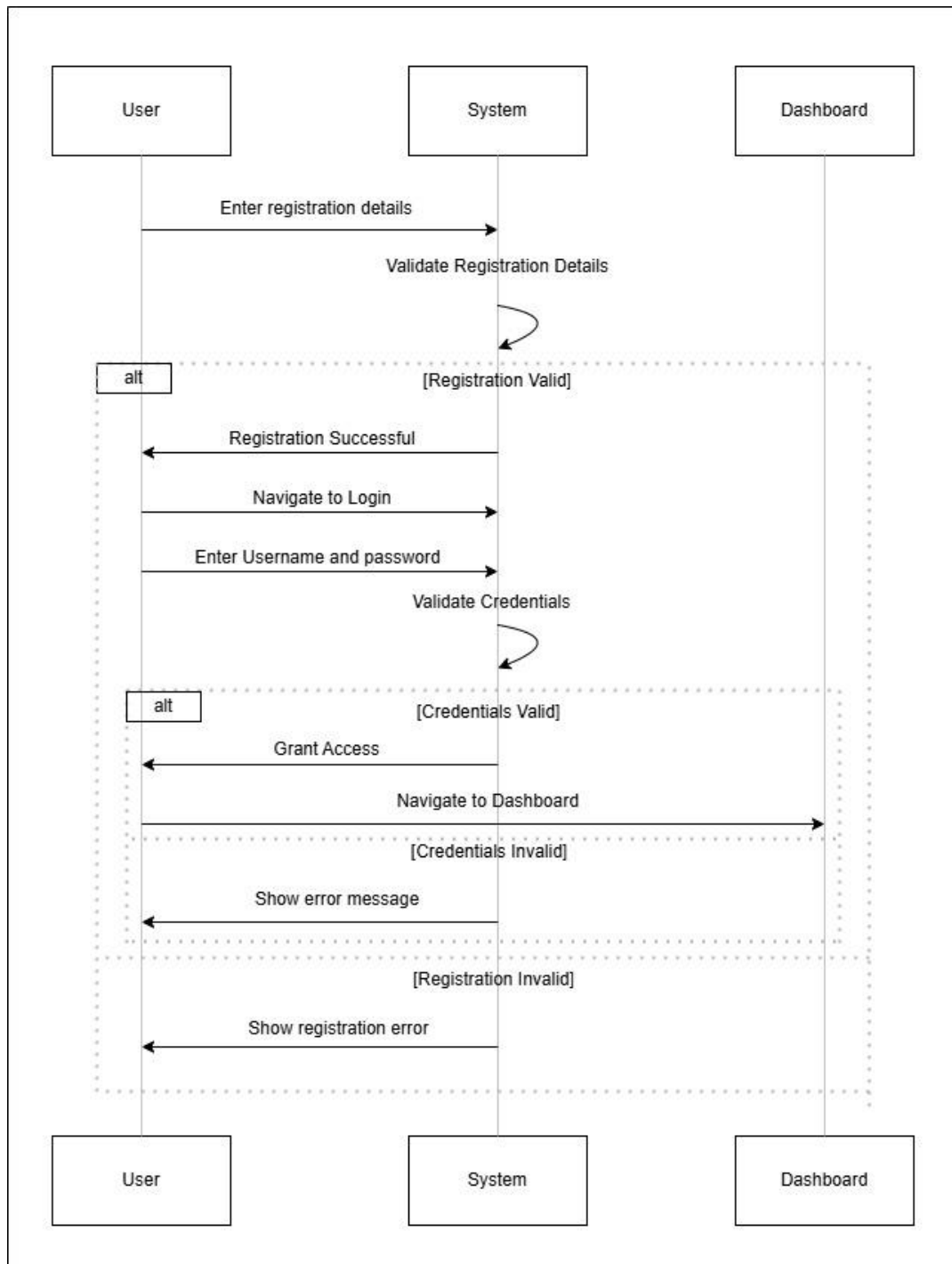


Figure 7. The sequence diagram of the login/logout

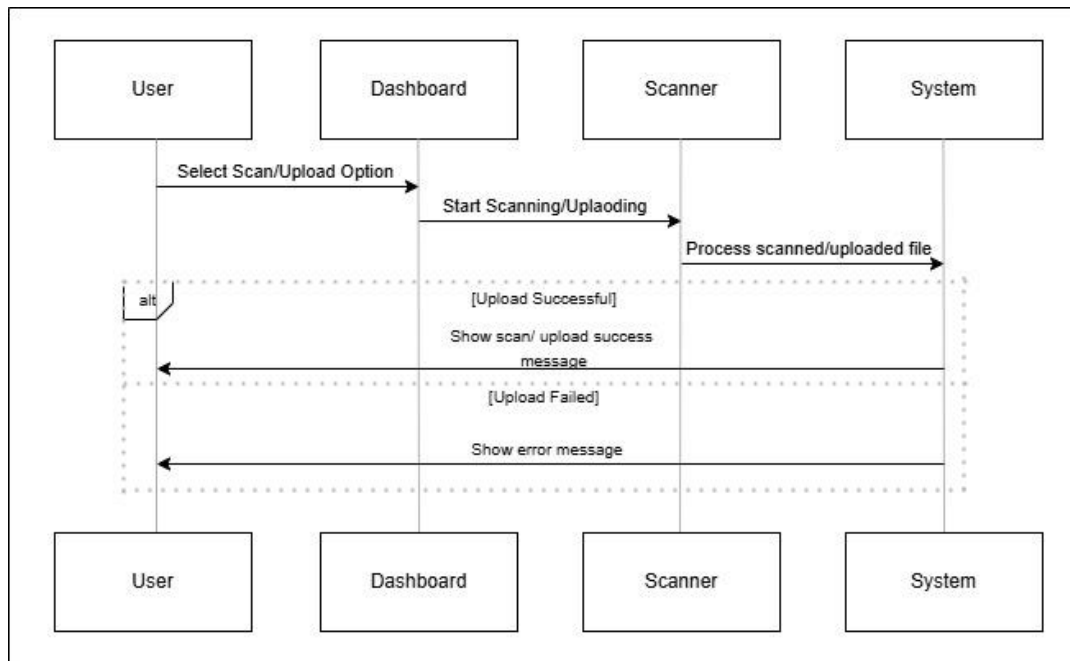


Figure 8. The sequence diagram of scan/upload

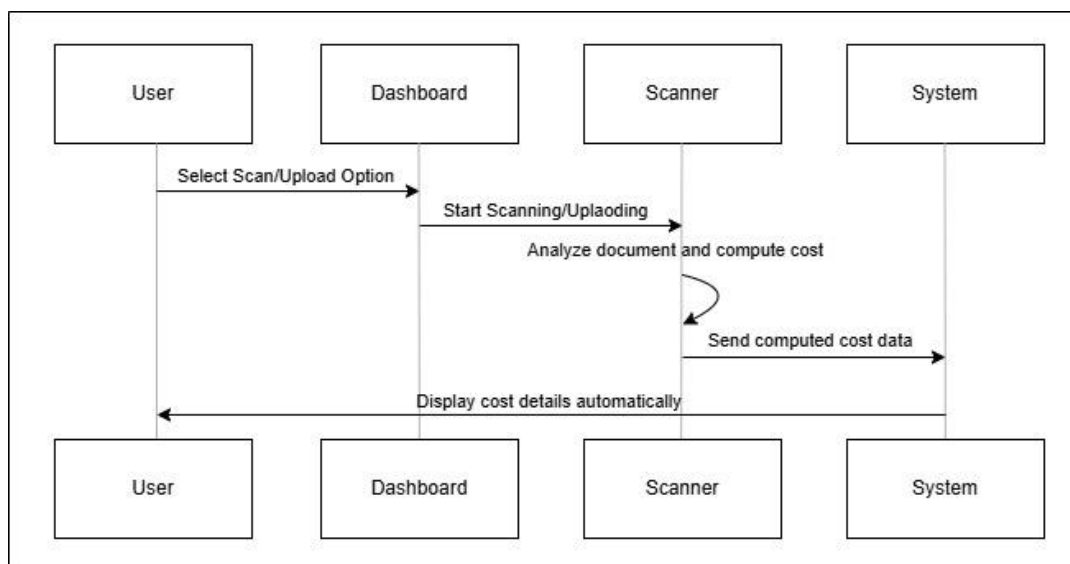


Figure 9. The sequence diagram of the View Costs

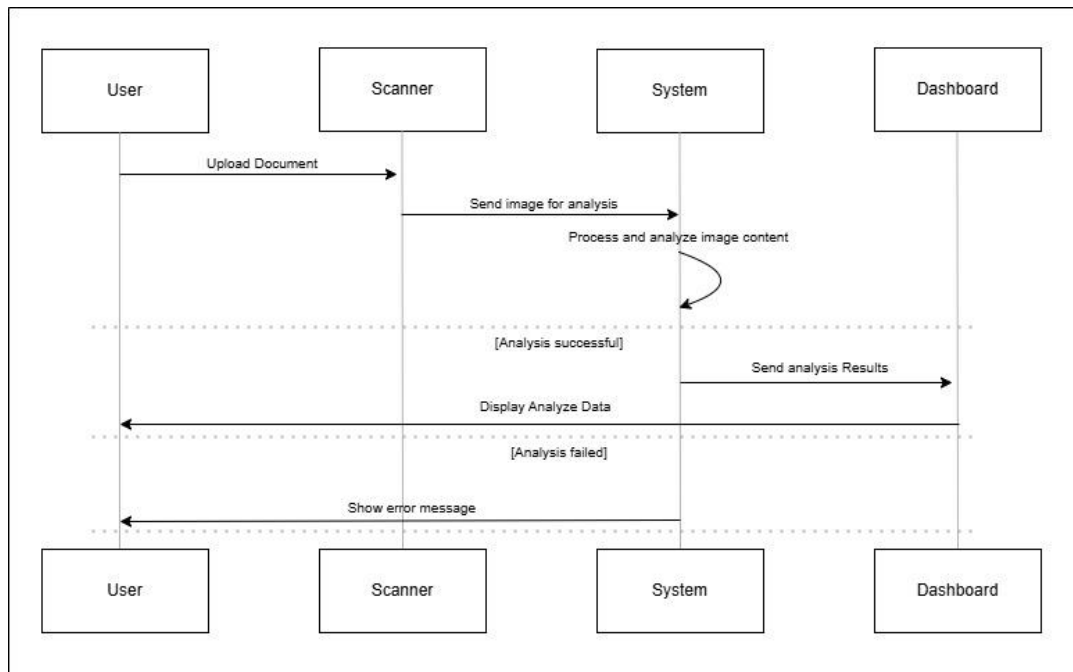


Figure 10. The sequence diagram of Analyze Image

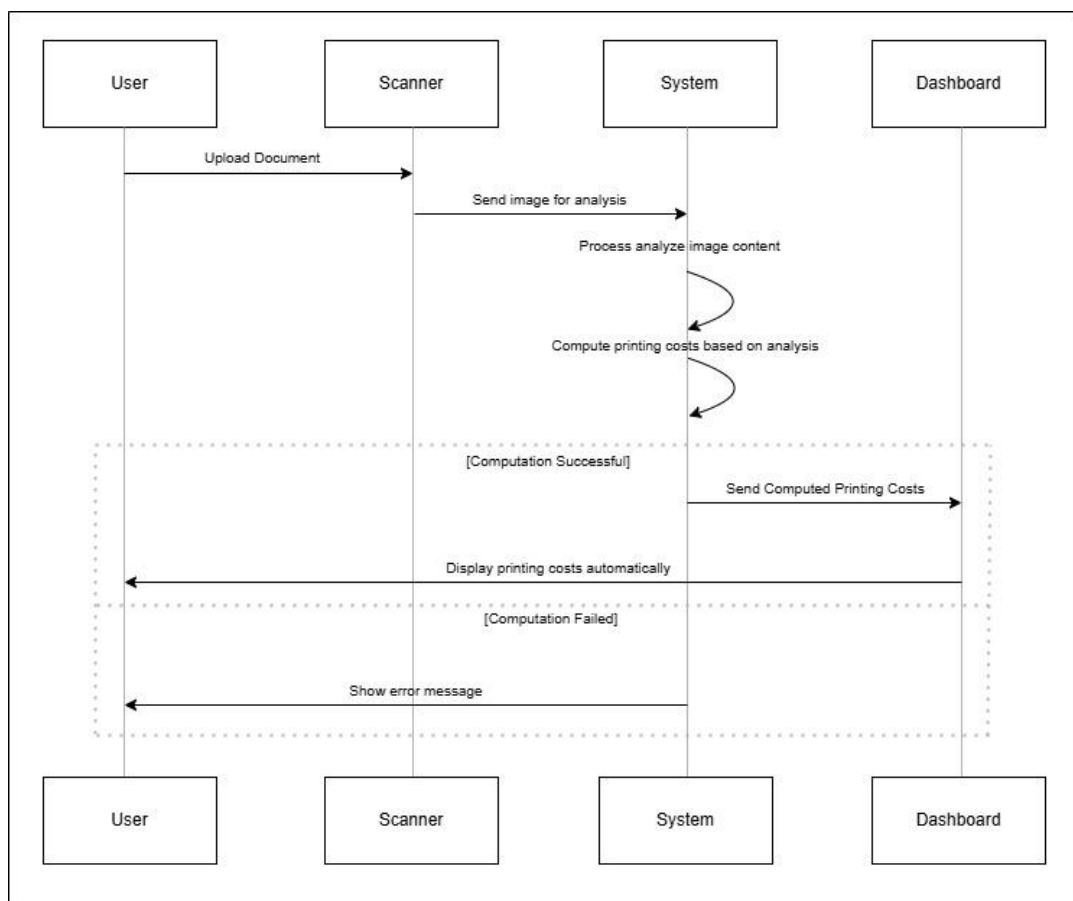


Figure 11. The sequence diagram of Compute Printing Costs

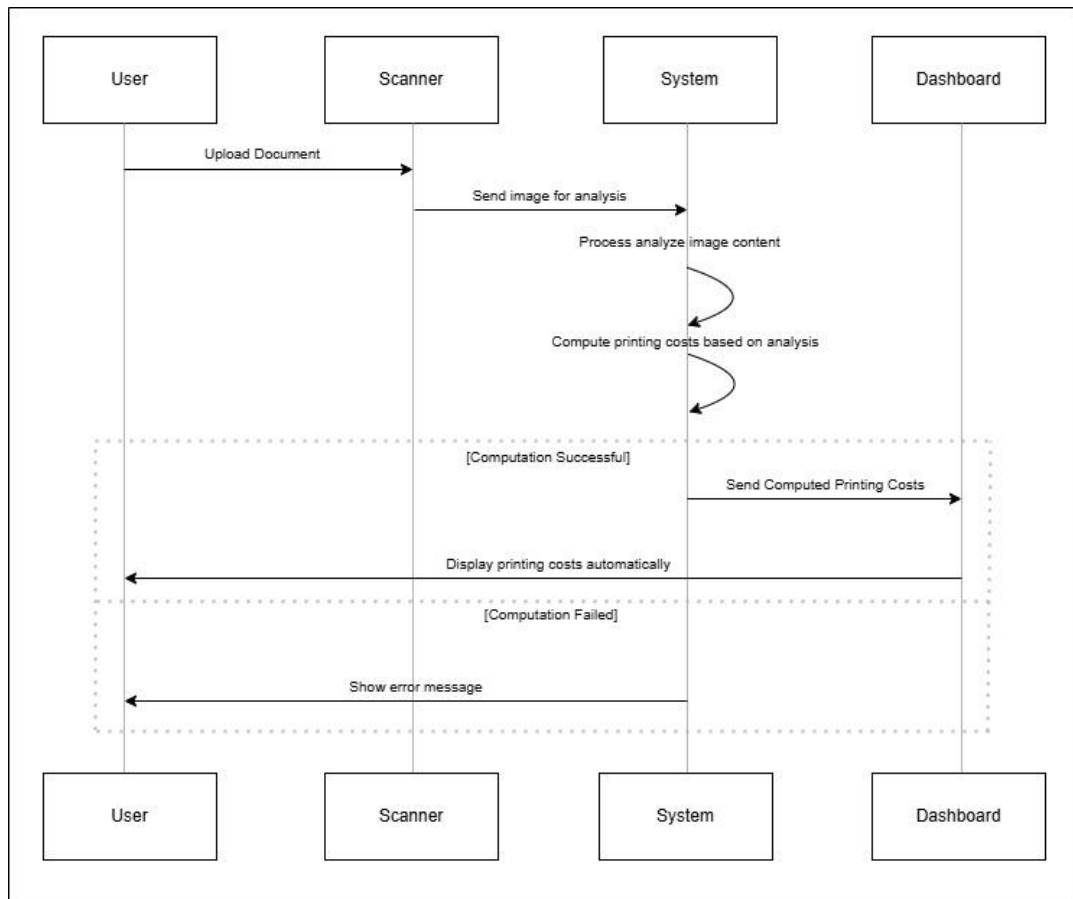


Figure 12. The sequence diagram of Sore Transaction Data

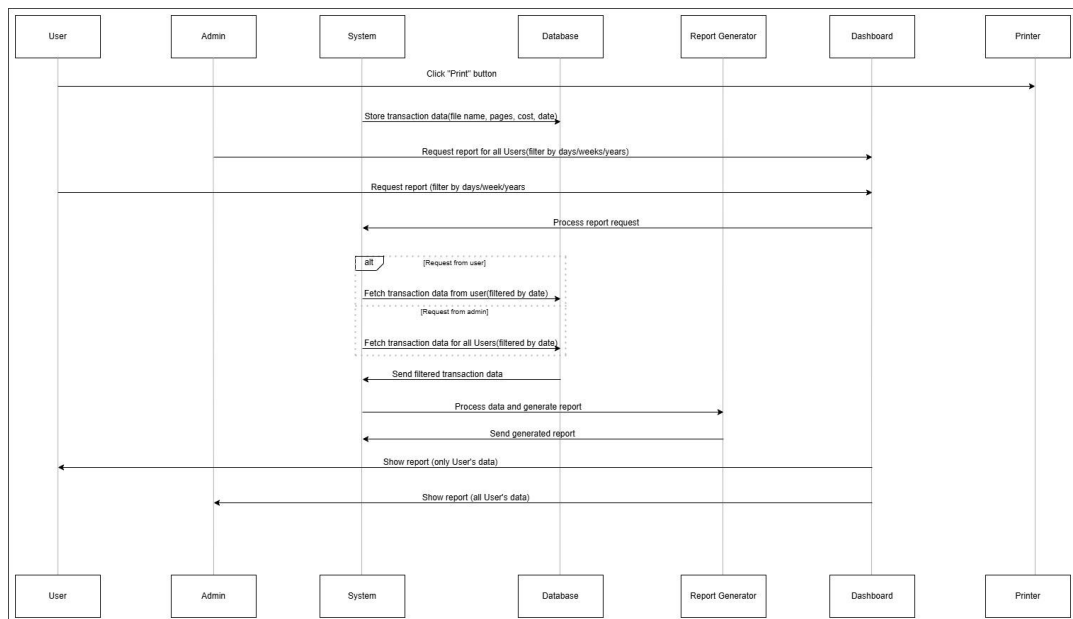


Figure 13. The sequence diagram of Generate Report

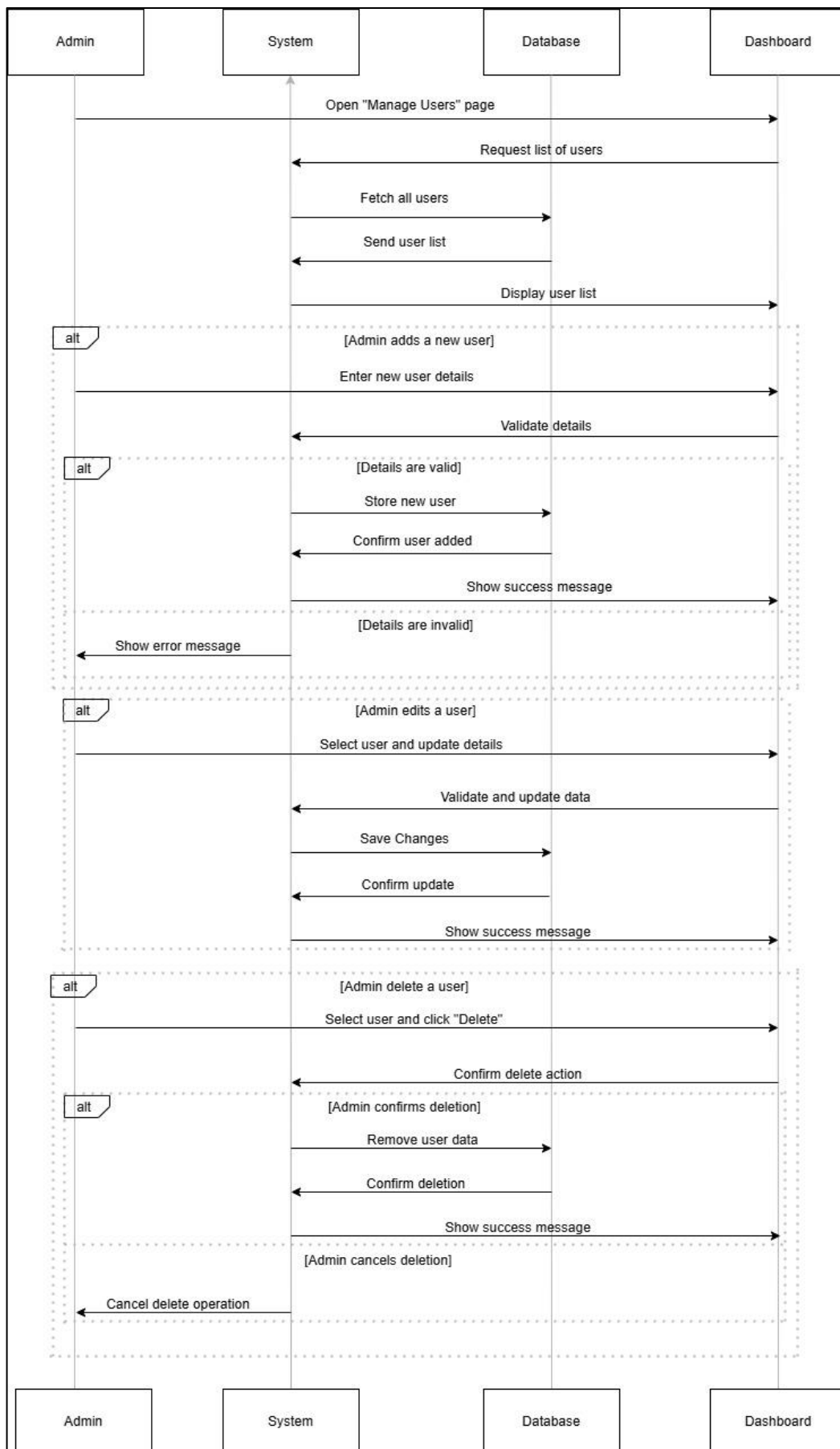


Figure 14. The sequence diagram of Manage Users

The above is a Sequence Diagram of Optimizing Printing and Scanning Cost Using Pixel-Based Algorithms. It is an interaction among three: User, System, and Admin. It begins with the user logging into the system, posting a document, and the system saving the file. The system then applies a Pixel-Based Algorithm to the document, performs text detection via OCR, and calculates the cost by covering text/image. The amount is then displayed to the user, who may accept or decline the print request. The system also has a transaction log; users and admins can view and create scanned document reports. The chart also illustrates the system's sequential document processing and cost estimation flow.

Data Dictionary

A Data Dictionary maintains records of the system database schema, declaring each table field, data types, and constraints.

Table 21. Data Dictionary

Table Name	Field Name	Data Type	Description
Users	user_id (PK)	INT (AUTO)	Unique user ID
	name	VARCHAR(100)	User full name
	email	VARCHAR(100)	Email
	user_role	ENUM('Admin,' 'User')	References user role
	password	VARCHAR(100)	Encrypted password
Documents	doc_id (PK)	INT (AUTO)	Unique Document ID
	user_id (FK)	INT	Foreign key to the Users table
	file_name	VARCHAR(255)	Name of the uploaded file
Analysis	upload_date	DATETIME	Date of upload
	analysis_id (PK)	INT (AUTO)	Distinct analysis ID
	doc_id (FK)	INT	Foreign key to the Documents table

	color_ratio	FLOAT	The color ratio in the document
	pages	INT	Number of pages in analysis
Cost Estimation	cost_id (PK)	INT (AUTO)	Distinct cost estimation ID
	analysis_id (FK)	INT	Foreign key to the Analysis table
	total_cost	DECIMAL(10,2)	Estimated cost based on color usage

Structuring and Quality Control

Table 22. Structuring and Quality Control

Aspects	Quality Control Measures
Functionality	Conduct unit, integration, and system testing.
Performance	Process massive documents efficiently in the system.
Security	Impose user authentication & data encryption.
User Experience	User testing with feedback.

System Documentation

The following shall be covered in the system documentation:

- User Manual - Step-by-step instructions on software usage.
- Technical Documentation - API documentation, system architecture, and coding framework.
- Installation Guide - Step-by-step guide to installation of the system.

System Installation & Configuration

- Installation Steps: Install software prerequisites (Python, Pillow/OpenCV).
- Install MySQL/Xampp database to track expenses.
- Make server and frontend compatible.

- System Start-Up & Administration
- First System Launch
- Final system testing. Final test the system.
- Train users on features.
- Release the system for live use.
- System Maintenance & Updates
- Continuous bug fixing and security updates.
- Deploy new features based on feedback.
- Reporting Module: Produces summary reports for cost tracking. Structuring and Quality Control

System Documentation

System documentation will comprise the following:

User Guide - User guide on utilizing the software.

Technical Documentation - API guide, system architecture, and coding design.

Installation Guide - System installation step-by-step.

- System Installation & Configuration
- Installation Steps: Install software dependencies (Python, Pillow/OpenCV).
- Install MySQL/Xampp database for cost

management.

- Make frontend and server compatible.

System Start-Up & Administration

First System Launch

- Perform final system testing.
- Train end-users for use.
- Release the system for live usage.

System Maintenance & Updates

- Scheduled bug fixes and security fixes.
- Implement new features based on feedback.
- Customer documents are uploaded to determine color usage and cost estimation.
- The system will read documents through a pixel-based algorithm to estimate color usage and cost.

Performer Included:

User, System, Cost Estimation Module

Steps

- User uploads file.
- The system checks the file format and saves the files.
- Pixel-Based Algorithm
- The algorithm checks color intensity.
- The cost estimation module computes the cost of the detected color.
- The system shows the user the cost estimation.

Development Approach

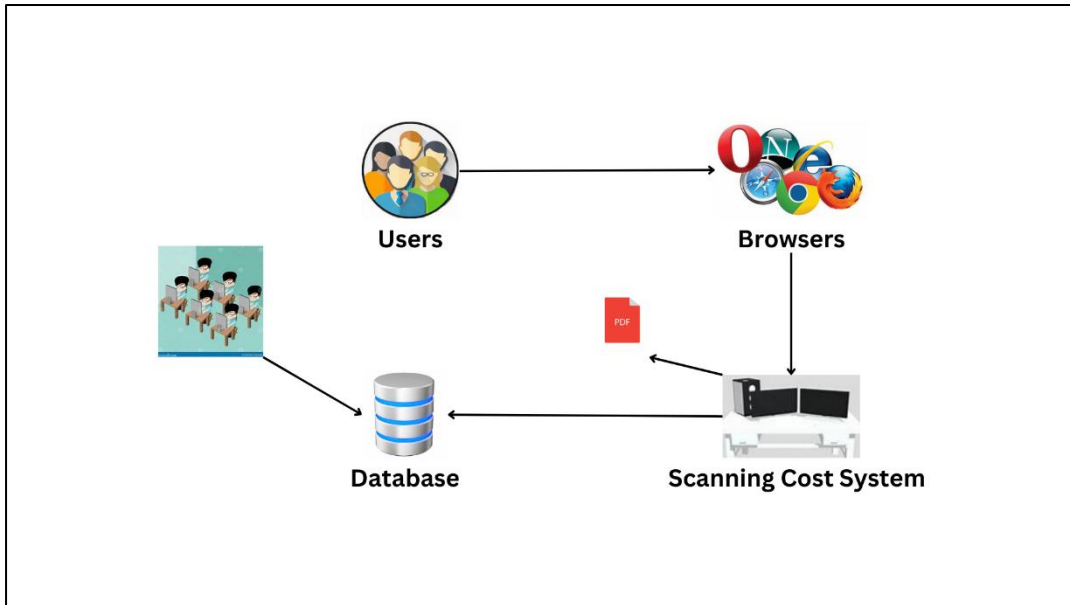


Figure 15. Web-Based Deployment Diagram of the System

The figure illustrates a Scanning Cost System Architecture, where users must interact with the system via web browsers to estimate scanning costs. Users upload documents via browsers, which are scanned by the Scanning Cost System to examine text-to-image ratios and provide accurate scanning costs. The system fetches and stores related data into a database in order to facilitate cost estimation efficiently. In addition, PDF files are passed from the scanning system and the database for processing. This setup maximizes resource allocation, reduces wasteful costs, and enhances the precision of cost estimates.

Frontend (CSS & HTML) and backend (Django) settings and API implementations to process data appropriately and finish the process. The most crucial operations are:

1. User creation and authentication

within the database.

2. Image processing is handled with Pillow and OpenCV.
3. Django REST Framework

is utilized to set up an API endpoint.

4. Making a quick operational interface is dealt with Bootstrap.
5. Uploading documents, report generation, and print functionality are set up.

Deployment Diagram

A three-tier architecture is implemented to deploy the system:

1. Client Layer: The user accesses the system through a web browser.
2. Requests are processed at the
3. application layer by the Django backend.
4. Database Layer: Data is stored and fetched from the MySQL database.

Deployment options:

- Local Server (deployed on XAMPP for testing).
- Cloud Hosting (Live Deployment Hostinger).

Testing

Tests executed on the system are:

- Testing individual modules, i.e., API functions, is called unit testing.
- Integration testing is used to test the data transfer between modules.
- Functional Testing: All system functions should be wanted.
- System response time and

performance is tested.

Validation, Verification, and Testing Plan

Table 23. Validation and Testing Plan

Phase	Objective	Testing Method
Verification	All modules should be ensured to be correct	User acceptance (UAT)
Static check and code inspection.	The system should be ensured to be as per the requirements.	testing
Verification		
Units testing	individual components.	Automated testing tools (e.g., pytest).
Integration Testing	Testing module-to-module.	API testing using Postman.
	Deployment testing how well and efficiently the system is working.	JMeter loading tests.

Installation of Database and MySQL include

1. d in the deployment.
2. Running an app on a host server or local server.
3. Set-up of environmental variables (e.g., API keys).
4. Last test for running smoothly.

5. Deployment for actual usage.

Maintenance

Long-term stability and performance are ensured through maintenance:

- Bug Fixes: Fixing user-reported issues.
- Feature enhancements: Introduction of additional features as requested by customers.
- Security patches: The process of encryption

and checking is being modified.

- Speed and scalability of the process are core pillars of performance tuning.

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